

CHE 205, Fall 2019, Problem Set 9

Instructor: Shafigh Mehraeen

Due date: 11:59pm, Wed, 11/27/2019

All problems are from the textbook by V.L. Law.

Submit two spreadsheets, one for each problem, on blackboard.

Problem 1 (10 pts). Rework problem 1 in problem set 8 using "method of lines" (FTCS), and the VBA implemented in example 8.5 in the book. Note that you have to modify this VBA script.

Problem 2 (10 pts). Exercise 8.5. Discard the last part of the problem, which is experimentation with different SOR factors and comparison of the results.

- Set guess values $\phi(x, y)=1$, Tolerance=1E-5, $\Delta x=\Delta y=0.1$, $0 \leq y \leq 1$, and $0 \leq x \leq 1$.
- Print out number of iterations as well as the maximum error.
- Plot $\phi(x, y = 0.5)$ against x and $\phi(x = 0.5, y)$ against y .

Exercise 8.5: Resolve the example elliptic PDE of Equation 8.23 using the following boundary conditions and with an SOR factor of 1.4. Use a grid such that $\Delta x = \Delta y$.

$$d\varphi(0, y)/dy = 0$$

Insulated left boundary

$$\varphi(1, y) = 1$$

Right side at 1

$$d\varphi(x, 0)/dx = 0$$

Insulated bottom boundary

$$\varphi(x, 1) = 0$$

Top side at 0

Experiment with different SOR factors and compare the results.